Linear Algebra II

Backpaper Examination

Instructions: All questions carry ten marks. Vector spaces are assumed to be finite dimensional.

1. The charactaeristic polynomial of the matrix below is $X^3 - 4X - 1$. Determine the missing entries.
$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 0 \\ 1 & * & * \end{bmatrix}$$

2. Let B be a real square matrix such that $B^2 = B$. Show that the dimension of the eigenspace with eigen value 1 equals the rank of B.

3. Let T_1, T_2 be linear operators on a vector space V over a field F such that $T_1 \circ T_2 = T_2 \circ T_1$. Let V_λ denote the subspace of V consisting of vectors v such that $T_1(v) = \lambda v$. Prove that V_λ is T_2-invariant. If $V_\lambda \neq \{0\}$, then will it have to contain eigen vectors for T_2? Justify your answer.

4. Prove that the eigenvalues of a Hermition matrix are real.

5. Let M be a symmetric real matrix of order n. Let $v \in \mathbb{R}^n$ be such that $M^2 v = 0$. Prove that $Mv = 0$.

6. Prove that a real symmetric matrix is positive definite if and only if its eigenvalues are positive.

7. Let P be a real sqaure matrix of order n such that $P^k = 0$ for some k. Then prove that $P^n = 0$.